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(54) **OBJECT ATTRIBUTE-BASED
WATERMARKING METHOD FOR
PREVENTING LEAKAGE OF DIGITAL
CONTENT**

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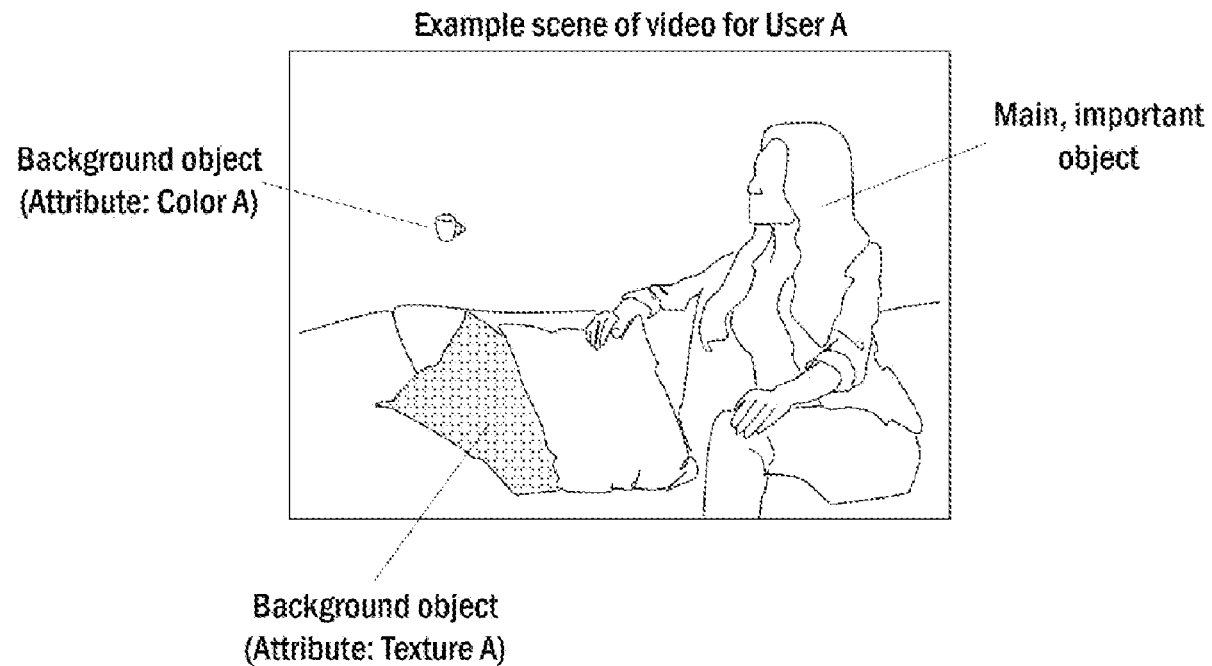
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(57) **ABSTRACT**

There is provided an object attribute-based watermarking method for preventing leakage of a digital content. A video watermarking method according to an embodiment may acquire a requested video and watermarking information upon receiving a request for video streaming from a user, may determine some of objects appearing in the video according to the watermarking information, may transform attributes of the determined objects, and may stream the transformed video. Accordingly, by watermarking an attribute of an object appearing in a video differently according to a user and providing the video, a watermark may be difficult to perceive, robustness against various attacks may be guaranteed, and a leakage path of a content may be checked.



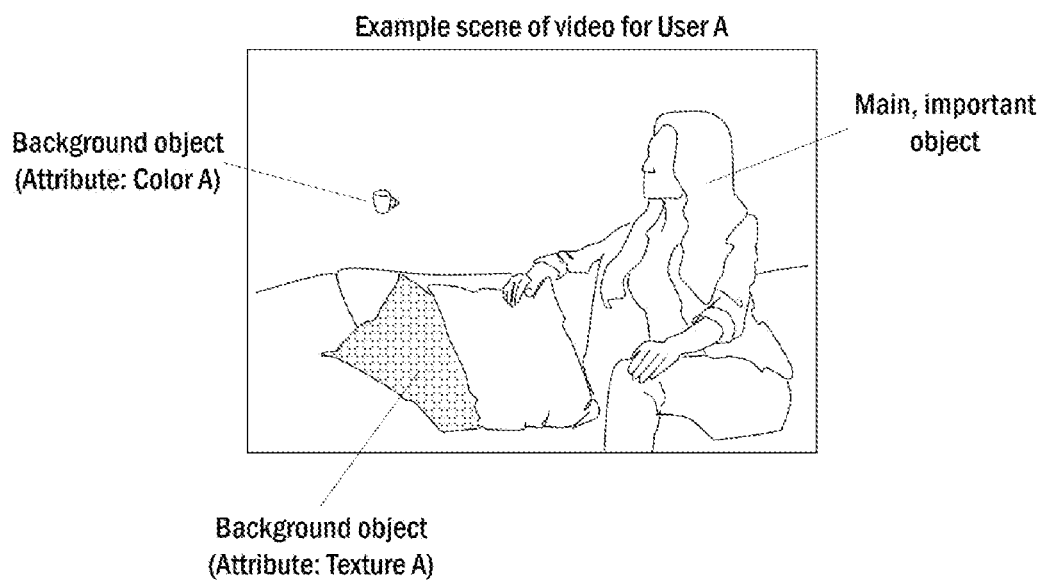


FIG. 1

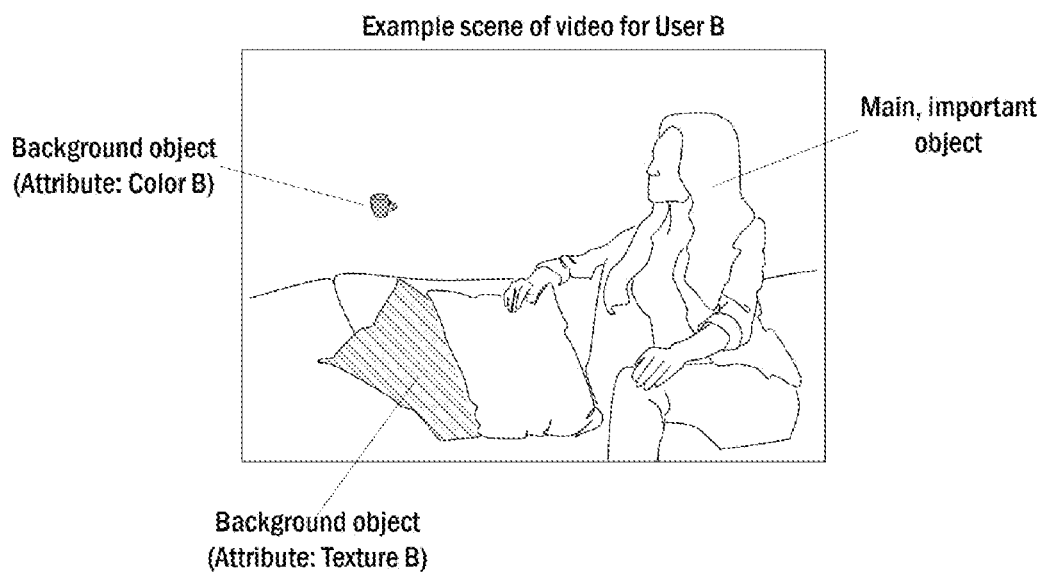
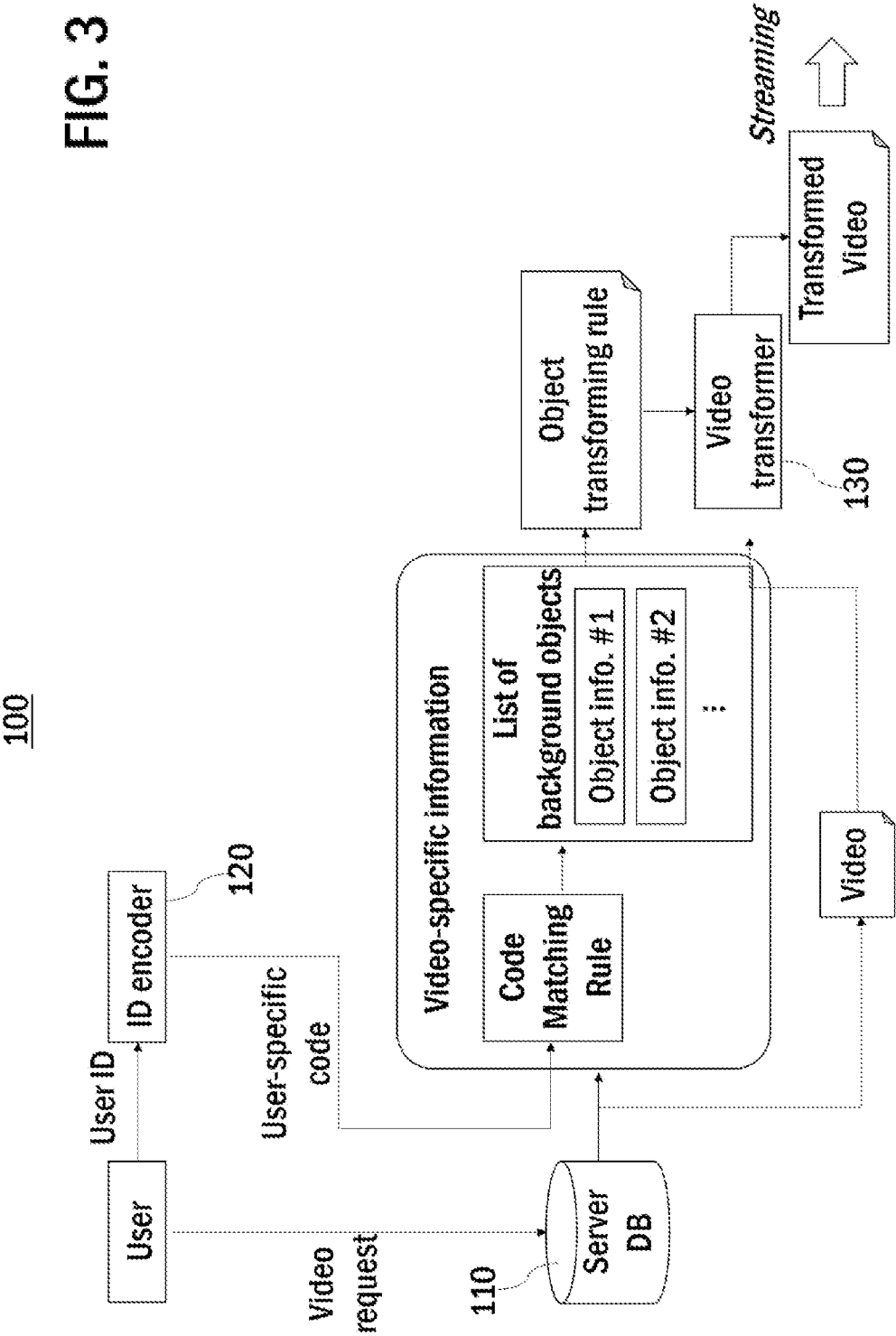


FIG. 2



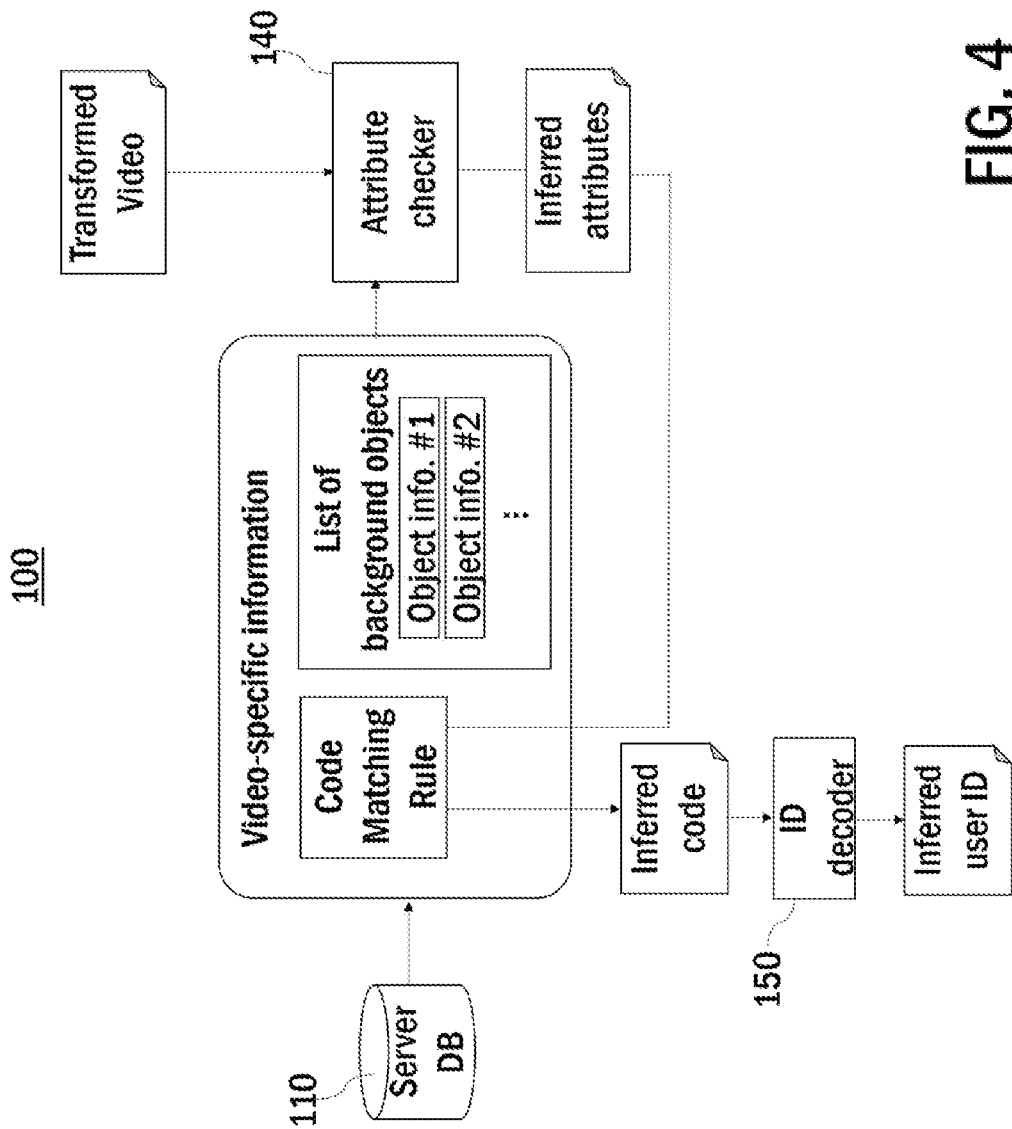


FIG. 4

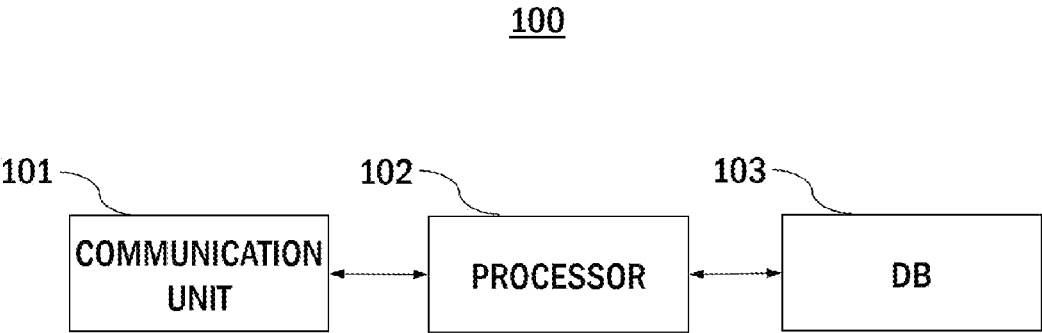


FIG. 5

**OBJECT ATTRIBUTE-BASED
WATERMARKING METHOD FOR
PREVENTING LEAKAGE OF DIGITAL
CONTENT**

**CROSS-REFERENCE TO RELATED
APPLICATION(S) AND CLAIM OF PRIORITY**

[0001] This application is based on and claims priority under 35 U.S.C. § 119 to Korean Patent Application No. 10-2023-0066130, filed on May 23, 2023, in the Korean Intellectual Property Office, the disclosure of which is herein incorporated by reference in its entirety.

BACKGROUND

Field

[0002] The disclosure relates to a digital content work protection technology, and more particularly, to a watermarking method for preventing leakage of a digital content and tracing a leakage path when the content is leaked.

Description of Related Art

[0003] Watermark technology for preventing leakage of contents such as digital cinema, video, or the like has developed various methods to protect content copyrights, and in particular, copyright protection technology is developing to be able to provide a service to a user through an imperceptible watermarking technique of disabling perception of a watermark, and simultaneously, to grasp a content leakage path.

[0004] Such watermarking technology should be able to authenticate against various attacks and should have robustness to have integrity. However, related-art technology employs a method of implanting a user-specific code by utilizing various characteristics of an image, and there may be techniques for having robustness, but the related-art technology may have a disadvantage of being vulnerable to new temporal, spatial attacks.

SUMMARY

[0005] The disclosure has been developed in order to solve the above-described problems, and an object of the disclosure is to provide an imperceptible watermarking method which is capable of grasping a video leakage path based on attributes of objects appearing in a video.

[0006] According to an embodiment of the disclosure to achieve the above-described object, a video watermarking method may include: receiving a request for video streaming from a user; acquiring the requested video and watermarking information; determining some of objects appearing in the video according to the watermarking information; transforming attributes of the determined objects; and streaming the transformed video.

[0007] The objects may be background objects.

[0008] Determining may include determining some background objects based on a user-specific code. Determining may include determining a background object to transform from the user-specific code based on a code matching rule in which a background object to transform is matched with a user-specific code.

[0009] Transforming may include transforming at least one of a color and a texture. A type of an attribute to be

transformed in a background object may be determined based on the user-specific code.

[0010] The video watermarking method according to the disclosure may further include generating a user-specific code by encoding a user ID, and determining may include using the generated user-specific code.

[0011] The video watermarking method according to the disclosure may further include: acquiring watermarking information of a leaked video; inferring a user-specific code applied to transformation of an attribute of a background object appearing in the leaked video, based on the watermarking information; and acquiring a user ID by decoding the inferred user-specific code.

[0012] Inferring the user-specific code may include: determining some background objects of which attributes are transformed in the leaked video, with reference to the watermarking information; and inferring the user-specific code from the result of determining with reference to a code matching rule.

[0013] According to another aspect of the disclosure, there is provided a video streaming server including: a communication unit configured to receive a request for video streaming from a user, and to stream the requested video; a DB configured to store videos and watermarking information; and a processor configured to acquire the requested video and watermarking information from the DB, to determine some of objects appearing in the video according to the watermarking information, to transform attributes of the determined objects, and to stream the transformed video through the communication unit.

[0014] According to still another aspect of the disclosure, there is provided a method for tracing a video leakage path, the method including: acquiring watermarking information of a leaked video; determining a transformed background object in the leaked video with reference to the watermarking information; inferring a user-specific code from the result of determining; and acquiring a user ID by decoding the inferred user-specific code.

[0015] According to yet another aspect of the disclosure, there is provided a video leakage path tracing system including: a DB configured to store videos and watermarking information; and a processor configured to acquire a leaked video and watermarking information, to determine a transformed background object in the leaked video with reference to the acquired watermarking information, to infer a user-specific code from the result of determining, and to acquire a user ID by decoding the inferred user-specific code.

[0016] According to embodiments of the disclosure as described above, by watermarking an attribute of an object appearing in a video differently according to a user and providing the video, a watermark may be difficult to perceive, robustness against various attacks may be guaranteed, and a leakage path of a content may be checked.

[0017] Other aspects, advantages, and salient features of the invention will become apparent to those skilled in the art from the following detailed description, which, taken in conjunction with the annexed drawings, discloses exemplary embodiments of the invention.

[0018] Before undertaking the DETAILED DESCRIPTION OF THE INVENTION below, it may be advantageous to set forth definitions of certain words and phrases used throughout this patent document: the terms “include” and “comprise,” as well as derivatives thereof, mean inclusion

without limitation; the term “or,” is inclusive, meaning and/or; the phrases “associated with” and “associated therewith,” as well as derivatives thereof, may mean to include, be included within, interconnect with, contain, be contained within, connect to or with, couple to or with, be communicable with, cooperate with, interleave, juxtapose, be proximate to, be bound to or with, have, have a property of, or the like. Definitions for certain words and phrases are provided throughout this patent document, those of ordinary skill in the art should understand that in many, if not most instances, such definitions apply to prior, as well as future uses of such defined words and phrases.

BRIEF DESCRIPTION OF THE DRAWINGS

[0019] For a more complete understanding of the present disclosure and its advantages, reference is now made to the following description taken in conjunction with the accompanying drawings, in which like reference numerals represent like parts:

[0020] FIG. 1 is a view provided to explain a concept of a video watermarking method according to an embodiment of the disclosure;

[0021] FIG. 2 is a view provided to explain the concept of the video watermarking method according to an embodiment of the disclosure;

[0022] FIG. 3 is a view illustrating a video watermarking method in a video streaming server according to another embodiment of the disclosure;

[0023] FIG. 4 is a view provide to explain a method of tracing a video leakage path in a video streaming server according to still another embodiment of the disclosure; and

[0024] FIG. 5 is a view illustrating a hardware configuration of a video streaming server according to yet another embodiment of the disclosure.

DETAILED DESCRIPTION

[0025] Hereinafter, the disclosure will be described in more detail with reference to the accompanying drawings.

[0026] There is provided an object attribute-based watermarking method for preventing leakage of a digital video content. The disclosure provides a technology for watermarking an attribute of a background object appearing in a video differently according to a user, and streaming data, so that the watermark is difficult to perceive, robustness against various attacks is guaranteed, and a content leakage path can be checked.

[0027] FIG. 1 is a view provided to explain a concept of a video watermarking method according to an embodiment of the disclosure, and FIG. 2 is a view provided to explain the concept of the video watermarking method according to an embodiment. The method according to an embodiment may watermark by transforming an attribute of a background object, which is an unimportant part, rather than a foreground object which is an important object noticed by a viewer in a video content, differently according to a user (viewer).

[0028] For example, even when a user A and a user B view the same scene of a video, the scene that the user A is viewing (shown in FIG. 1) and the scene that the user B is viewing (shown in FIG. 2) may be different from each other. In this case, the different parts may be attributes (for example, color and texture in FIGS. 1, 2) of background

objects (cup and cushion in FIGS. 1, 2) that do not influence users' appreciation of the video and understanding of the story.

[0029] Since what is transformed in the video content is a background object, the users may view the video without perceiving a watermark applied to the video. In addition, since the position or shape of the background object is not changed and only the attribute such as color or texture is transformed, there is almost no damage being applied to the video content itself and robustness against various attacks is provided.

[0030] The watermark applied differently according to a user, that is, the attribute of the background object transformed differently according to a user, may be used for tracing a leakage path when the video content is leaked hereinafter, that is, a user who leaks the video.

[0031] FIG. 3 is a view illustrating a video watermarking method in a video streaming server according to another embodiment.

[0032] As shown in FIG. 3, when a user requests a video content that the user wishes to view from a streaming server 100, the streaming server 100 acquires the video requested by the user and information necessary for watermarking the corresponding video from a database (DB) 110.

[0033] The information necessary for watermarking (hereinafter, referred to as ‘watermarking information’) contains a code matching rule and a list of background objects, and may be defined differently according to a video.

[0034] The list of background objects contains information on background objects (Object info. #1, Object info. #2, . . .) that do not raise the matter even when they are transformed for watermarking in the video content. As described above, the background objects are objects that do not cause any inconvenience in appreciating the video and understanding the story even when they are transformed. The information on the background objects may include the following information:

[0035] 1) Information on a time at which a corresponding background object appears;

[0036] 2) Information on a position of a corresponding background object;

[0037] 3) Type (class) of a corresponding background object;

[0038] 4) Attribute information to be transformed for a corresponding background object.

[0039] The attribute to be transformed in the background object may include a color, a texture of the background object. The attribute needs to be defined to be transformed to have a much different value from an original attribute value so as to be robust against various attacks. For example, if the color of a corresponding background object is red, the color may be defined to be transformed to a much different color, such as blue, green, rather than a similar color such as pink.

[0040] An ID encoder 120 generates a user-specific code by encoding a user ID. The code matching rule is a rule for determining a background object to transform based on the user-specific code.

[0041] For example, if the user-specific code is a 4-digit number (abcd), the streaming server 100 may determine the ‘ab’-th background object information and the ‘cd’-th background object according to the code matching rule. Specifically, if the user-specific code is “0513”, information on the

05-th background object (Object info. #05) and information on the 13-th background object (Object info. #13) may be determined.

[0042] A video transformer **130** may watermark by transforming corresponding background objects in the video with reference to the determined information on the background object. Accordingly, a video in which some background objects are transformed may be streamed to a user. For natural transformation, the video transformer **130** may utilize a deep learning algorithm or an image processing technique.

[0043] FIG. 4 is a view provided to explain a method for tracing a video leakage path in the video streaming server **100** according to still another embodiment.

[0044] When a streamed video is illegally leaked, watermarking information of the leaked video is acquired from the DB **110** in order to trace a leakage path. By doing so, a code matching rule of the leaked video and a list of background objects are acquired.

[0045] Thereafter, an attribute checker **140** grasps attributes of the background objects in the background object list and checks whether the attributes are transformed, and determine background objects of which the attributes are transformed.

[0046] The streaming server **100** infers a user-specific code from the result of determining by the attribute checker **140** with reference to the code matching rule. For example, when the background objects of which the attributes are transformed are the 05-th background object and the 13-th background object, and the code matching rule provides that background object information corresponding to the first two digits and the last two digits in the user-specific code are determined, the streaming server **100** may infer the user-specific code as "0513".

[0047] Thereafter, an ID decoder **150** restores a user ID by decoding the inferred user-specific code. The restored user ID may be regarded as an ID of a user who leaks the video.

[0048] FIG. 5 is a view illustrating a hardware configuration of the video streaming server **100** according to yet another embodiment. The video streaming server **100** according to an embodiment may include a communication unit **101**, a processor **102**, and a DB **103** as shown in FIG. 5.

[0049] The communication unit **101** is a communication means for receiving a request of a user and streaming a video content requested by the user.

[0050] The processor **102** performs the watermarking process proposed in FIG. 3 and user ID encoding/decoding, video transformation (watermarking), attribute checking necessary for tracing a leakage path, proposed in FIG. 4.

[0051] The DB **103** is a storage in which videos and watermarking information are stored.

[0052] Up to now, the object attribute-based watermarking method for preventing leakage of digital contents has been described in detail with reference to preferred embodiments.

[0053] Embodiments of the disclosure propose a new watermarking method which provides a video content having a different object attribute according to a user, thereby making it difficult to perceive a watermark, achieving robustness against various attacks, and checking a leakage path of a content.

[0054] In the above-described embodiments, the background objects to be transformed are determined based on the user-specific code and the code matching rule. Changes

can be made thereto. For example, a type of an attribute to be transformed in a background object may be determined based on the user-specific code and the code matching rule. That is, only the color of a background object may be transformed or only the texture may be transformed according to the user-specific code, or both the color and the texture may be transformed.

[0055] The technical concept of the present disclosure may be applied to a computer-readable recording medium which records a computer program for performing the functions of the apparatus and the method according to the present embodiments. In addition, the technical idea according to various embodiments of the present disclosure may be implemented in the form of a computer readable code recorded on the computer-readable recording medium. The computer-readable recording medium may be any data storage device that can be read by a computer and can store data. For example, the computer-readable recording medium may be a read only memory (ROM), a random access memory (RAM), a CD-ROM, a magnetic tape, a floppy disk, an optical disk, a hard disk drive, or the like. A computer readable code or program that is stored in the computer readable recording medium may be transmitted via a network connected between computers.

[0056] In addition, while preferred embodiments of the present disclosure have been illustrated and described, the present disclosure is not limited to the above-described specific embodiments. Various changes can be made by a person skilled in the art without departing from the scope of the present disclosure claimed in claims, and also, changed embodiments should not be understood as being separate from the technical idea or prospect of the present disclosure.

What is claimed is:

1. A video watermarking method comprising:

receiving a request for video streaming from a user;
acquiring the requested video and watermarking information;

determining some of objects appearing in the video according to the watermarking information;
transforming attributes of the determined objects; and
streaming the transformed video.

2. The video watermarking method of claim 1, wherein the objects are background objects.

3. The video watermarking method of claim 2, wherein determining comprises determining some background objects based on a user-specific code.

4. The video watermarking method of claim 3, wherein determining comprises determining a background object to transform from the user-specific code based on a code matching rule in which a background object to transform is matched with a user-specific code.

5. The video watermarking method of claim 4, wherein transforming comprises transforming at least one of a color and a texture.

6. The video watermarking method of claim 5, wherein a type of an attribute to be transformed in a background object is determined based on the user-specific code.

7. The video watermarking method of claim 3, further comprising generating a user-specific code by encoding a user ID,

wherein determining comprises using the generated user-specific code.

8. The video watermarking method of claim 7, further comprising:

acquiring watermarking information of a leaked video;
inferring a user-specific code applied to transformation of
an attribute of a background object appearing in the
leaked video, based on the watermarking information;
and
acquiring a user ID by decoding the inferred user-specific
code.

9. The video watermarking method of claim 8, wherein
inferring the user-specific code comprises:
determining some background objects of which attributes
are transformed in the leaked video, with reference to
the watermarking information; and
inferring the user-specific code from the result of deter-
mining with reference to a code matching rule.

10. A video streaming server comprising:
a communication unit configured to receive a request for
video streaming from a user, and to stream the
requested video;

a DB configured to store videos and watermarking infor-
mation; and
a processor configured to acquire the requested video and
watermarking information from the DB, to determine
some of objects appearing in the video according to the
watermarking information, to transform attributes of
the determined objects, and to stream the transformed
video through the communication unit.

11. A method for tracing a video leakage path, the method
comprising:
acquiring watermarking information of a leaked video;
determining a transformed background object in the
leaked video with reference to the watermarking infor-
mation;
inferring a user-specific code from the result of determin-
ing; and
acquiring a user ID by decoding the inferred user-specific
code.

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